

DSAI 150: Foundations of Data Science

Fall 2026 | Section 501 | MWF 10:20–11:10 AM, Room 2005

Instructor: Dr. Elena Ruiz | Office hours: Tue 2–4 PM | (Sample syllabus created for the SyllaSync demo. Fictional course.)

Course Description

An introduction to the data science workflow: collecting, cleaning, visualizing and modeling data with Python. Students complete weekly programming homework, in-lab exercises, two evening midterms, a team final project, and a cumulative final exam. Classes begin Monday, August 24, 2026. Week 1 is the week of August 24.

Grading

Component	Weight	Details
Homework	20%	8 assignments, due Fridays 11:59 PM. Lowest score dropped.
Quizzes	10%	5 in-class quizzes (2% each)
Labs	15%	6 lab reports (2.5% each), due Thursdays
Midterm Exam 1	15%	Evening exam
Midterm Exam 2	15%	Evening exam
Final Project	10%	Team project, report + presentation
Final Exam	15%	Cumulative

Grading scale: A 90–100, B 80–89.99, C 70–79.99, D 60–69.99, F below 60. No curve. Late homework loses 10% per day, up to 3 days.

Course Schedule

Week	Dates	Topic	Due / Assessments
1	Aug 24–28	Intro, Python refresher	
2	Aug 31–Sep 4	Data types & pandas	Lab 1 (Thu 9/3); HW 1 (Fri 9/4)
3	Sep 7–11	Data cleaning	Quiz 1 (Wed 9/9)
4	Sep 14–18	Exploratory analysis	Lab 2 (Thu 9/17); HW 2 (Fri 9/18)
5	Sep 21–25	Visualization	
6	Sep 28–Oct 2	Review; probability basics	MIDTERM 1: Tue 9/29, 7:00–9:00 PM; Quiz 2 (Wed 9/30); Lab 3 (Thu 10/1); HW 3 (Fri 10/2)
7	Oct 5–9	Sampling & distributions	
8	Oct 12–16	Hypothesis testing	Lab 4 (Thu 10/15); HW 4 (Fri 10/16)
9	Oct 19–23	Linear regression	Quiz 3 (Wed 10/21); Project proposal (ungraded) Fri 10/23
10	Oct 26–30	Classification	HW 5 (Fri 10/30)
11	Nov 2–6	Review; model evaluation	MIDTERM 2: Tue 11/3, 7:00–9:00 PM; Lab 5 (Thu 11/5)

Week	Dates	Topic	Due / Assessments
12	Nov 9–13	Clustering	Quiz 4 (Wed 11/11); HW 6 (Fri 11/13)
13	Nov 16–20	Text data	Lab 6 due Thursday of this week; HW 7 (Fri 11/20)
14	Nov 23–27	Ethics in data science	No class Nov 26–27 (Thanksgiving)
15	Nov 30–Dec 4	Project work	Quiz 5 (date TBA); HW 8 (Fri 12/4)
16	Dec 7–8	Presentations	Final Project report due Mon 12/7 11:59 PM; presentations Tue 12/8 in class
Finals	Dec 14		FINAL EXAM: Mon Dec 14, 10:30 AM–12:30 PM

Policies

Make-up exams require documentation submitted within 2 business days. Academic integrity rules apply to all work; collaboration on homework is allowed but code must be your own.